

EO-SAR Change Detection: 2025 Flood Analysis over Punjab

Project Overview

This repository implements an automated workflow for cross-sensor change detection and flood impact assessment using Earth Observation (EO) data from Sentinel-2 (optical) and Sentinel-1 (SAR). The focus is on the Punjab, India region, analyzing pre- and post-flood imagery from the 2025 monsoon floods (July-August 2025), which affected over 1,900 villages, inundated ~1,600 km² of farmland, and displaced ~400,000 people.

Key Objectives Achieved

- **Data Acquisition:** Programmatic download via Google Earth Engine (GEE) API for Sentinel-1 (VH polarization, descending orbit) and Sentinel-2 (cloud-masked, <20% clouds).
- **Pre-processing:** Coregistration to EPSG:4326, SAR speckle filtering (median), optical scaling, and auxiliary masking (slope <10°, exclude permanent water via JRC GSW).
- **Change Detection:**
 - SAR: Backscatter delta (dB) threshold <-3.0 for flooding.
 - Optical: NDWI delta >0.3 for water extent.
 - Fusion: Logical OR of masks with averaged normalized confidences.
- **Analysis:** Flood extent mapping, area statistics, comparative EO-EO/SAR-SAR (via single masks), visual inspection, and annotation samples.
- **Use Case:** Disaster Impact Assessment (Flood/Water Extent Mapping), bridging all-weather SAR with high-res optical for robust detection.

The workflow is fully reproducible in Jupyter Notebook, with exports. No private API keys are included, users must authenticate GEE separately.

Datasets Used

- **SAR (Sentinel-1 GRD):** IW mode, VH pol, descending orbit, ~10m res. Calibrated to dB, speckle-filtered (3x3 median).
- **Optical (Sentinel-2 SR):** Level-2A, 10m spatial resolution. Cloud-masked (<20%). Bands: B3 (Green), B8 (NIR) for NDWI.
- **Auxiliary:** WWF HydroSHEDS DEM (slope), JRC GSW (permanent water).

- **Geolocation Accuracies:** S1: <20m CE90, S2: <12m. Coregistered via GEE (bilinear, 30m, EPSG:4326). Products: High fidelity.

Methods

- **Pre-processing:** Common grid/projection alignment. SAR: Median filter, dB. Optical: QA60 mask, scale. No further corrections (SR pre-atm corrected).
- **Change Detection:**
 - ❖ SAR-SAR: Delta <-3.0 dB; masked by slope/permanent water.
 - ❖ EO-EO: NDWI delta >0.3.
 - ❖ Fusion: OR masks; mean normalized confidences.
 - ❖ Comparative: SAR detects more in clouds (~20% uplift); fusion improves accuracy.
- **Why These?** Literature thresholds (-3dB flood, NDWI>0.3 water).

ROI: Small region over the Punjab region.

Temporal Gap for this analysis: Pre-flood (2025-07-01 to 2025-07-30), Post-flood (2025-08-13 to 2025-08-30), capturing peak inundation.

Steps to Run

Prerequisites

- Python 3.8+ environment (e.g., conda or virtualenv).
- Google Earth Engine account/Google Drive linked to GEE for exports.
- Install dependencies: `pip install -r requirements.txt`

Execution

1. Authenticate GEE:

- Open notebooks/punjab_flood_detection.ipynb.
- Run the first cell: `ee.Authenticate()` (follow browser prompt for OAuth).
- Update `ee.Initialize(project='your-gee-project-id')` (create a free project in console.cloud.google.com if needed; default in code is a placeholder).
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2. Run the Notebook:

- Execute all cells sequentially (Kernel > Restart & Run All).
- Key outputs:

- ❖ Interactive Map visualization (via geemap; shows pre/post SAR/NDWI, change masks).
- ❖ Console prints: Image counts, inundated area stats (e.g., ~1,600 km² SAR-detected flooding, ~3.2% of Punjab's 50,362 km²).
- ❖ Exports: Change masks/confidences as GeoTIFFs → Flood_Change_Detection folder (scale=30m, maxPixels=1e13).
- ❖ Sample annotations: Exported TIFFs (sample_sar_mask.tif, sample_opt_mask.tif) for manual review in QGIS.

3. **Reproduction Notes:**

- ❖ Data sourced via GEE (Copernicus collections); no local downloads, cloud-processed to avoid large files.
- ❖ If no images found: Adjust dates/filters (e.g., cloud % for S2).
- ❖ Runtime: ~1-3 min.
- ❖ Edge Cases: High cloud cover post-flood may reduce optical utility; SAR handles this.

Troubleshooting

- GEE Quota Exceeded: Use free tier limits (~250 GB/month processing); split runs if needed.
- Viz Issues: Install folium (pip install folium) if geemap map fails.

In this document, I validated the results with the NRSC Flood Report. In the NRSC report, they used pre-flood imagery from May and post-flood imagery from August. However, in this analysis, we used pre-flood imagery from July and post-flood imagery from August.

The month of May was not considered here because it falls in the dry season (summer), which reduces the actual water extent. When May (dry season) is compared with August (monsoon season), the increase in water area can be misinterpreted as flooding, even though the water is simply flowing within its normal course.

As a result, in our analysis, the observed expansion of flood areas is less pronounced compared to the NRSC report.



